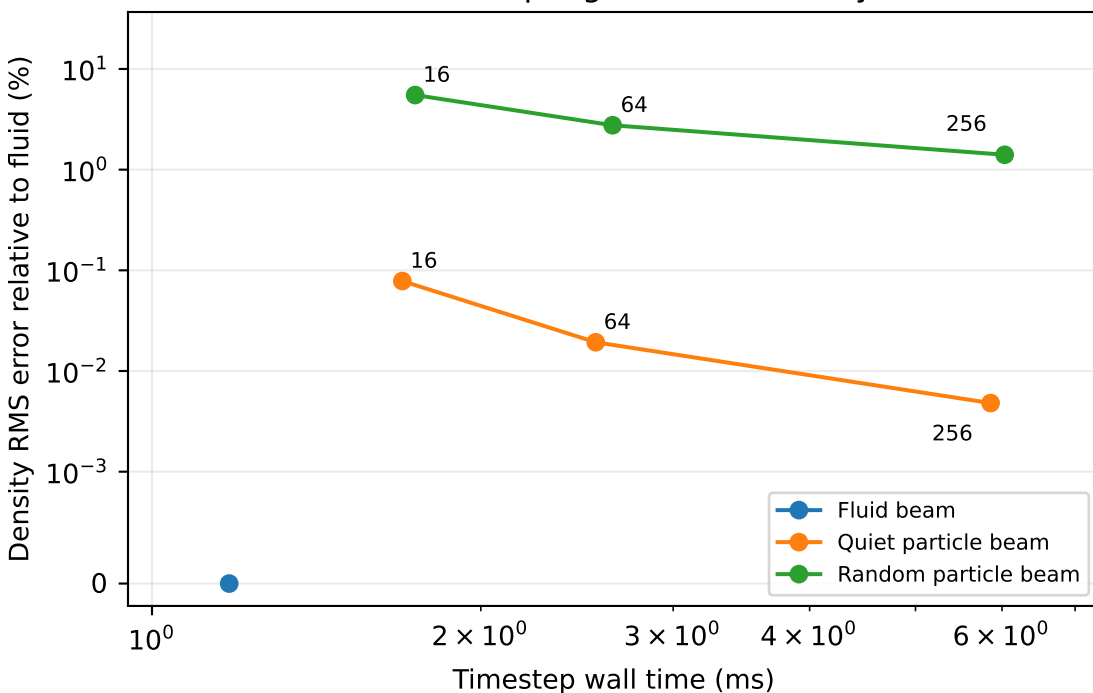
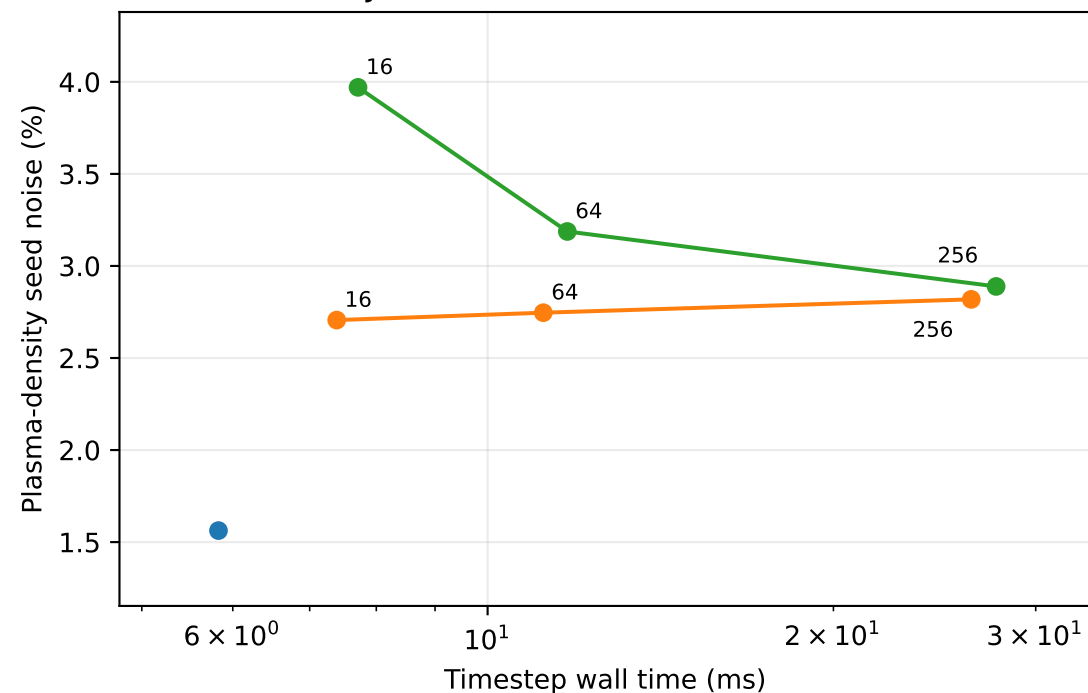


Perlmutter A100, RZ: measured noise and cost
Top: 64×256 , 6 seeds; bottom: 256×1024 , 3 seeds, quiet beam 64 particles/cell
Bars: 99% timing intervals; labels: beam particles/cell; synthetic MCC regression rates

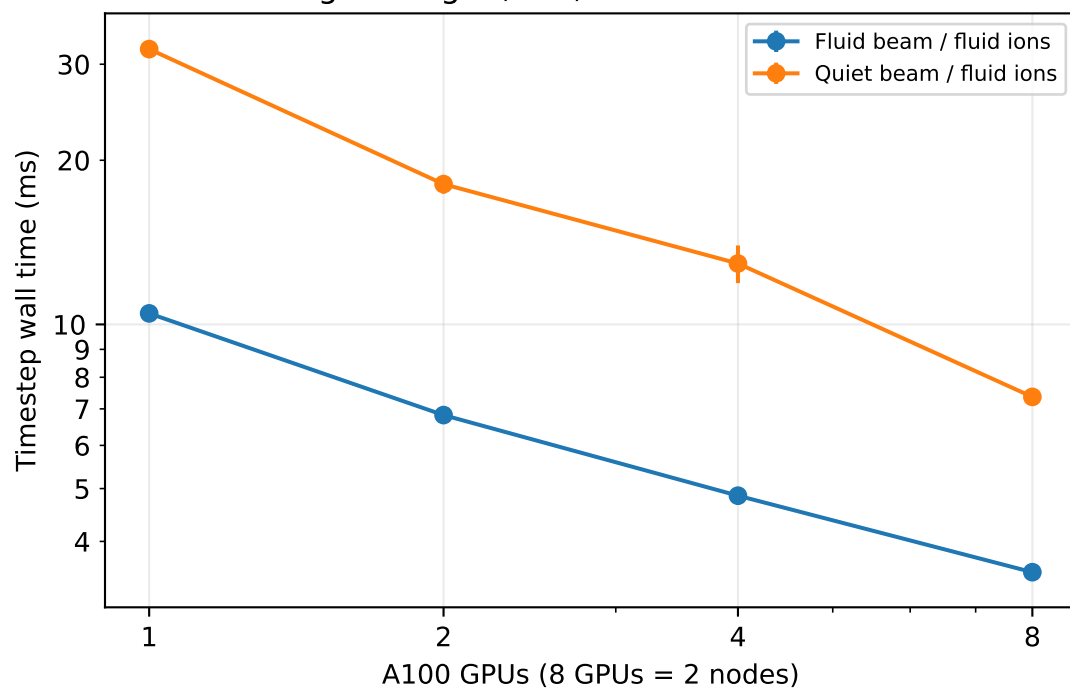
Beam sampling error: fields only



PJG + all electron MCC channels



Strong scaling: 1,048,576 identical electrons



Strong scaling: coupled production and MCC

